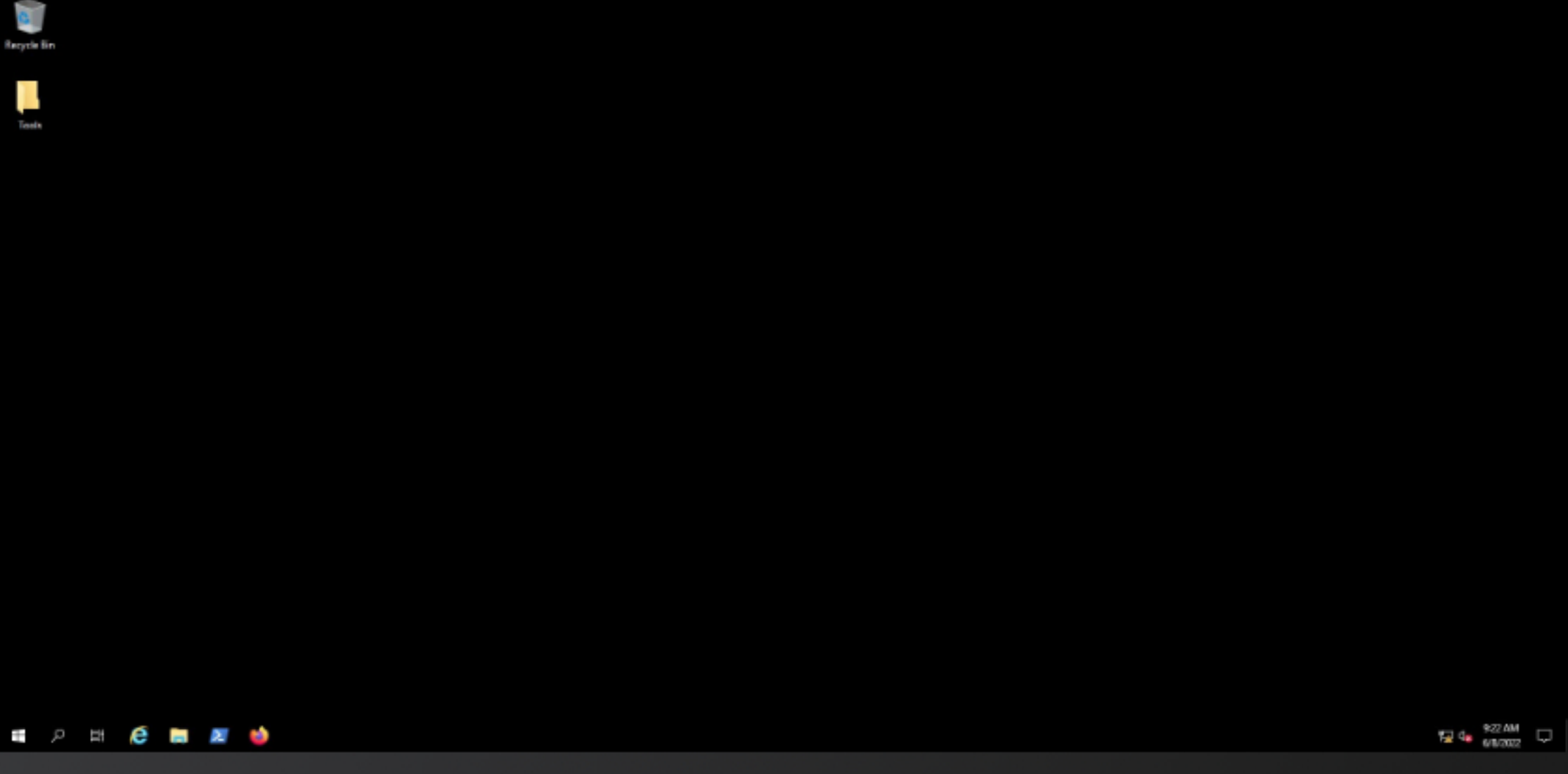


SOLUTIONS

Below, you can find solutions for each task. Remember though, that you can follow your own strategy, which may be different from the one explained in the following lab.

Windows Machine

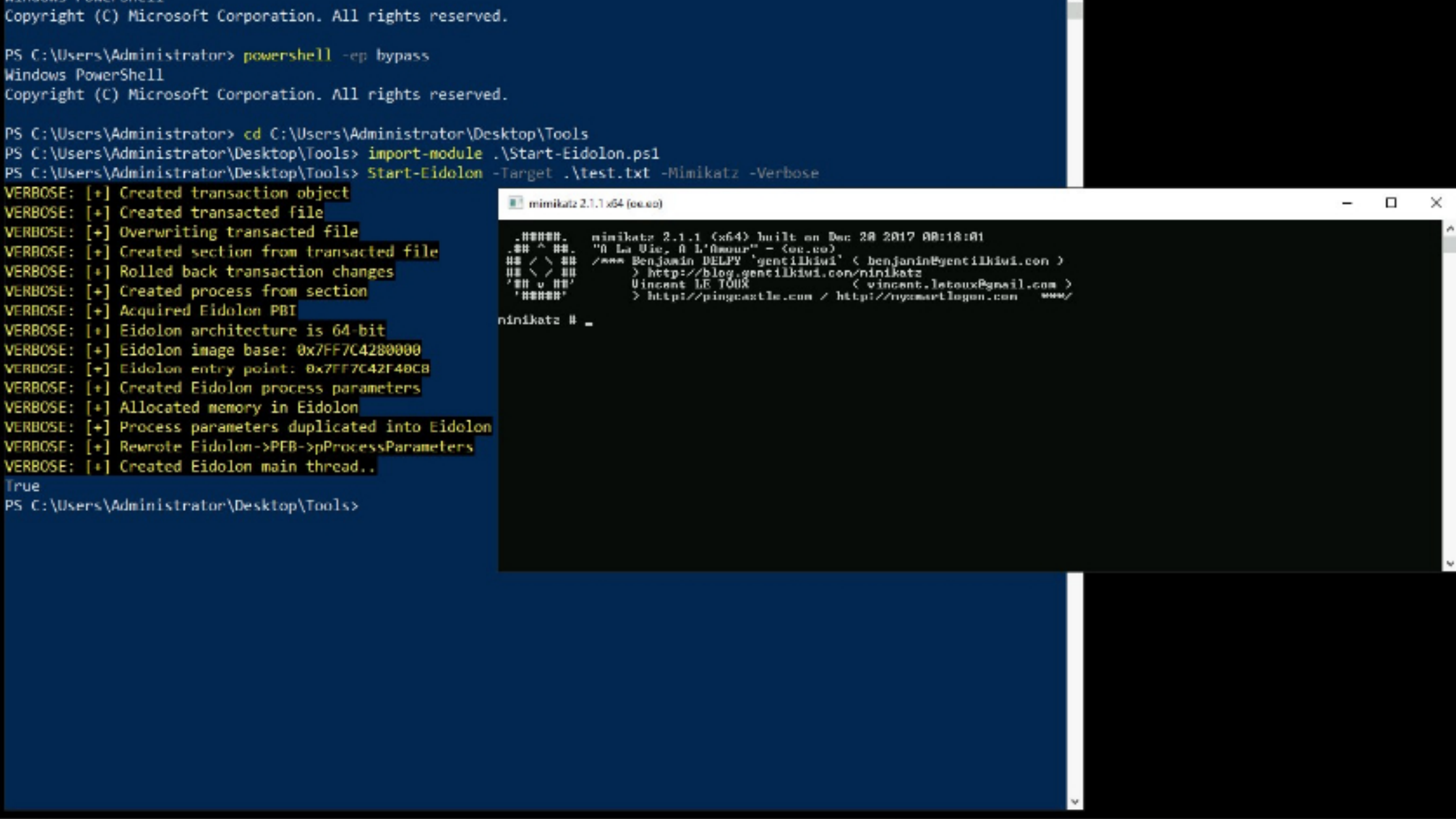


Task 1. Emulate and detect Process Doppelganging

To emulate a Process Doppelganging attack and then detect it using PE Sieve, perform the following.

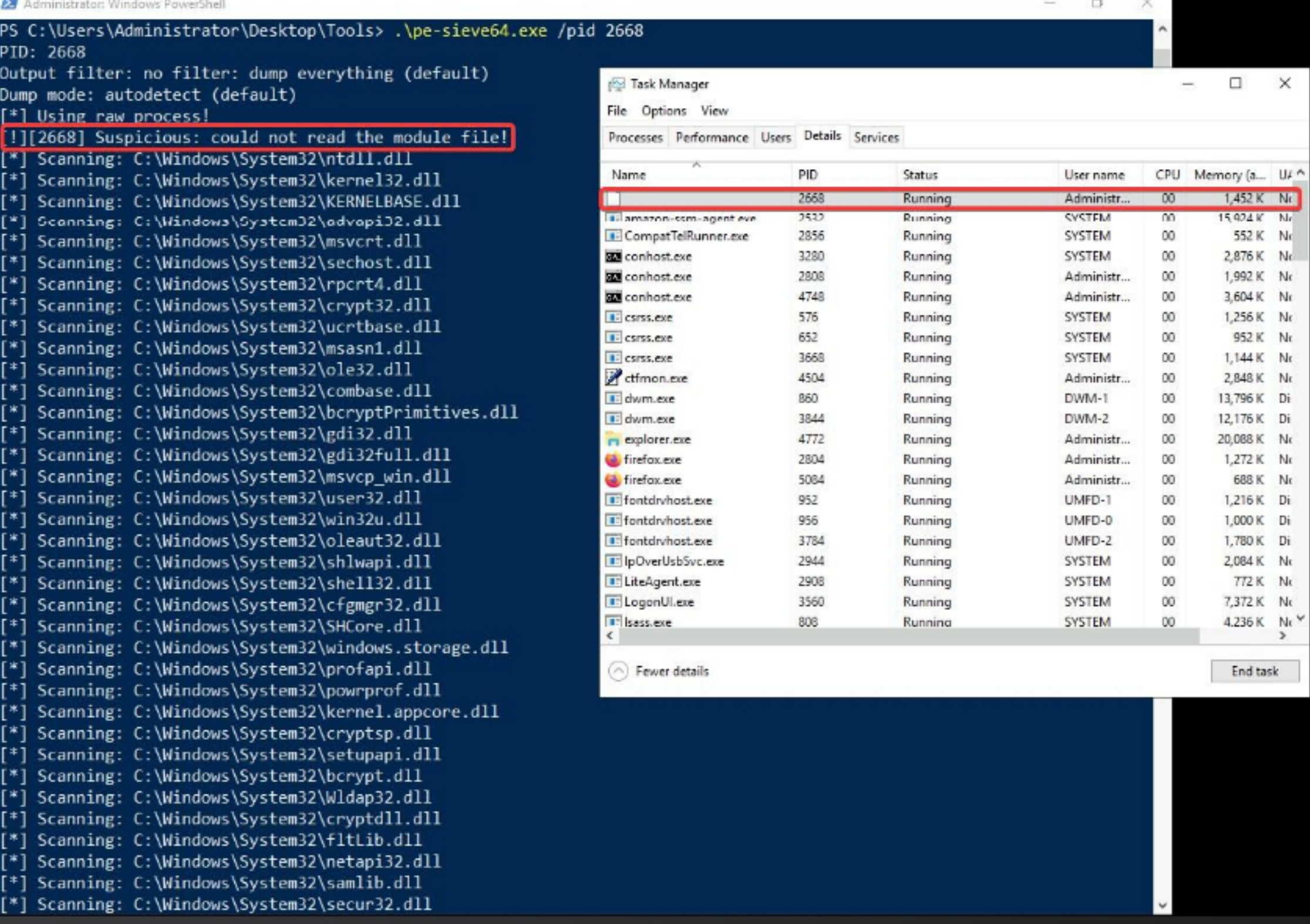
- Execute *powershell.exe* as an administrator and execute:

```
powershell -ep bypass
cd C:\Users\Administrator\Desktop\Tools
import-module .\Start-Bidolon.ps1
Start-Bidolon -Target .\test.txt -Mimikatz -Verbose
```

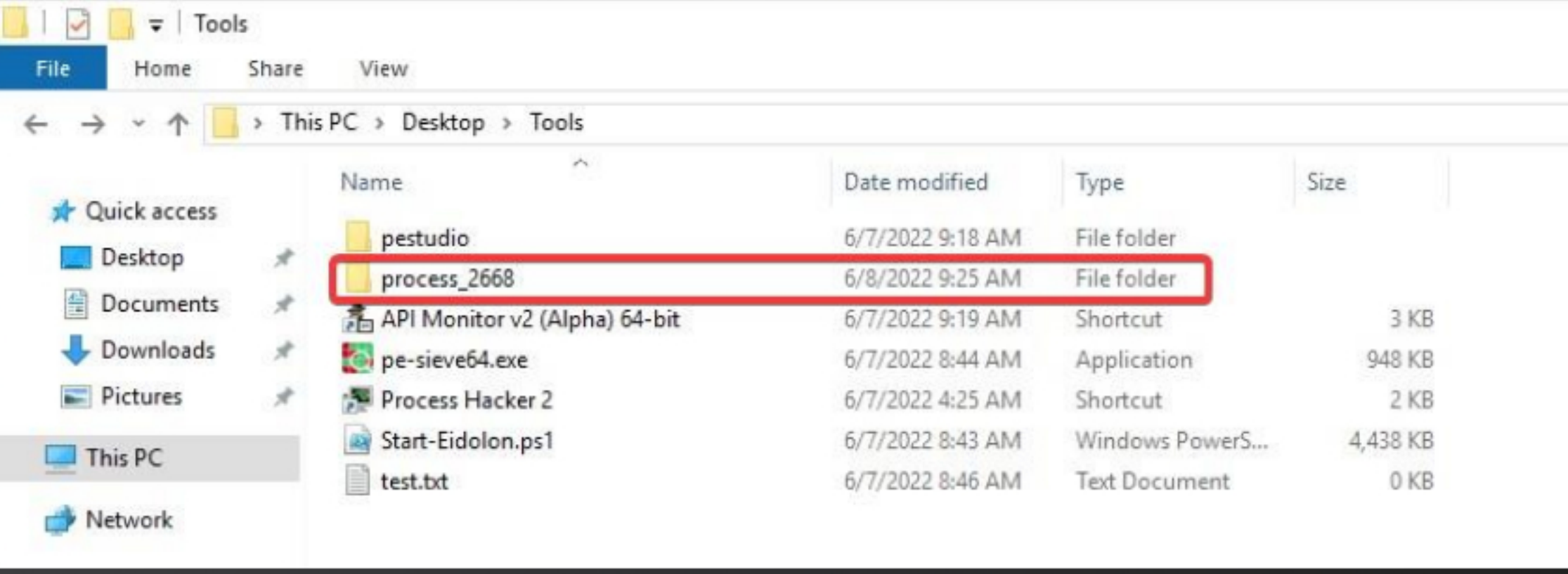


- Now identify the PID of the *test.txt* process and execute *cmd.exe* as an administrator. Then execute:

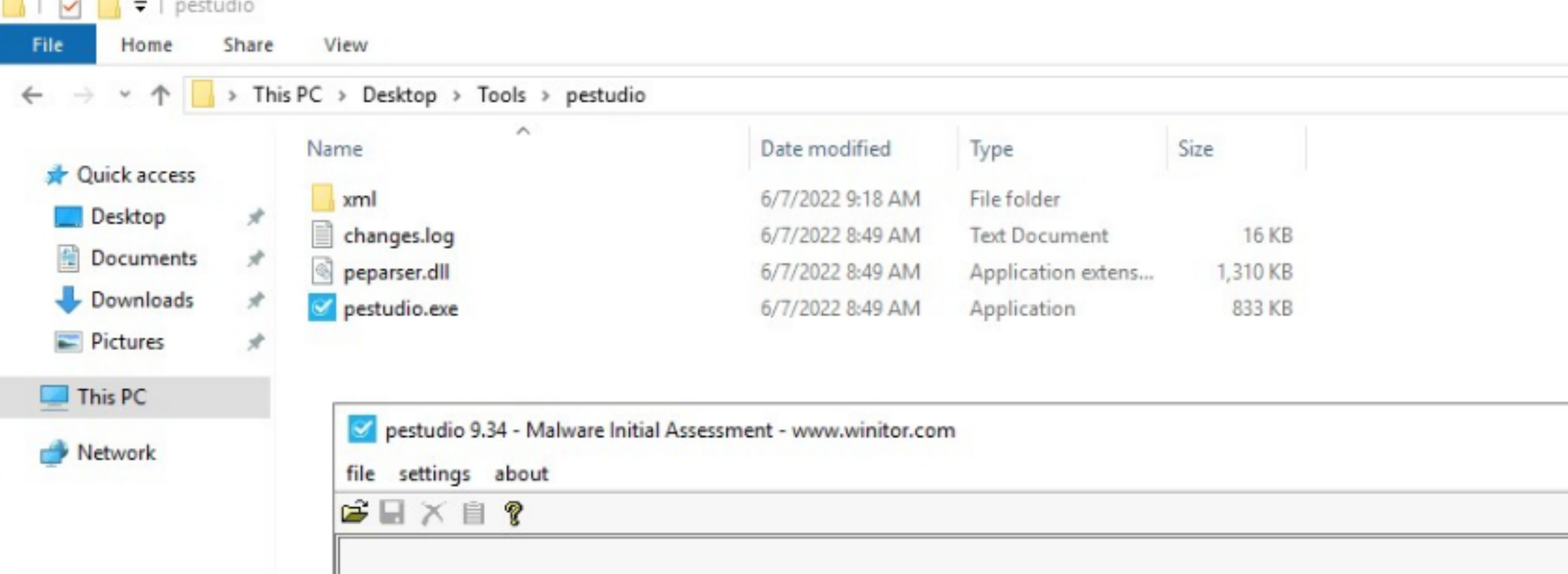
```
cd C:\Users\Administrator\Desktop\Tools
.\pe-sieve64.exe /pid PID_of_test_txt_process
```



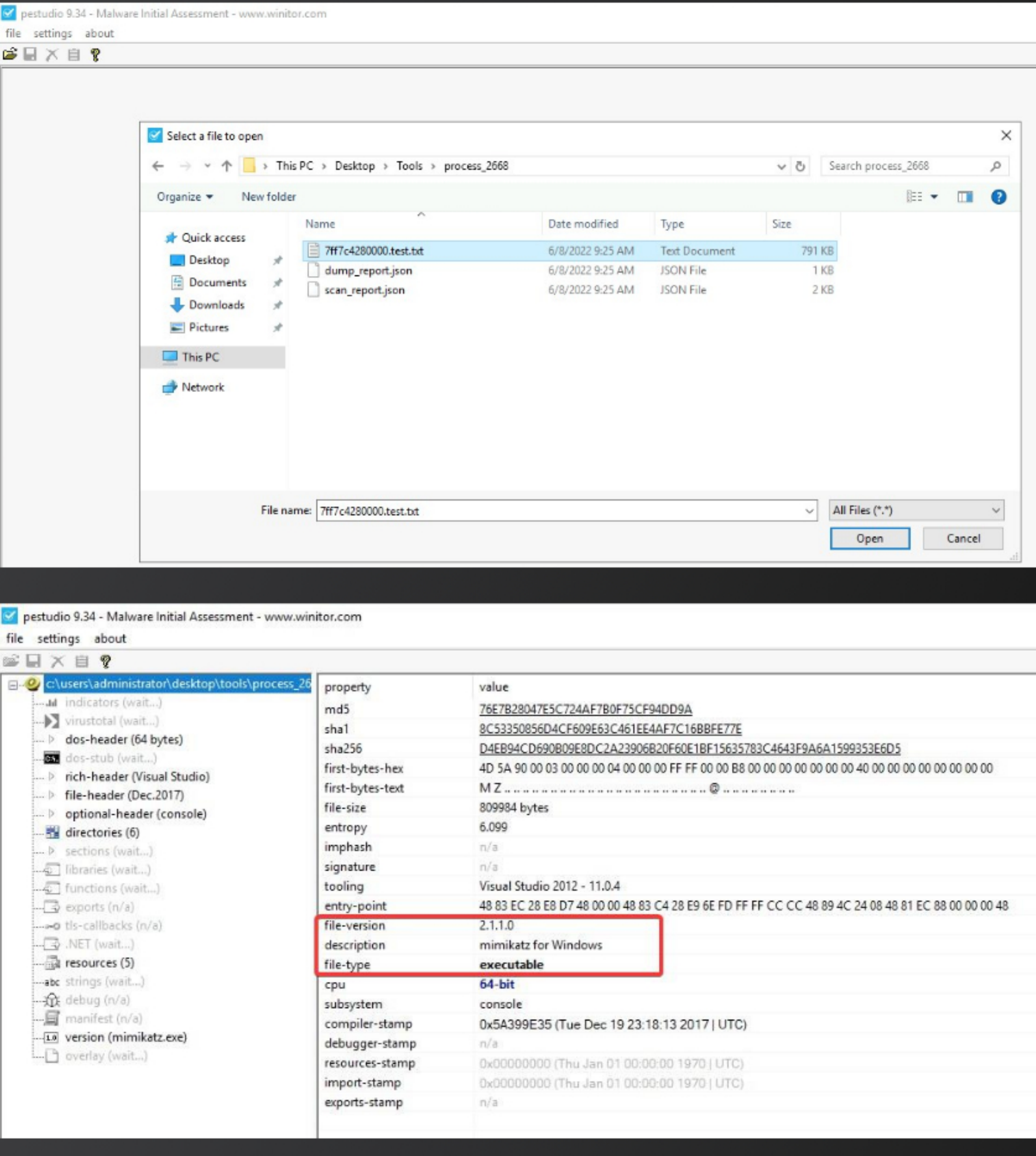
A folder named *process_PID* will be created inside the *C:\Users\Administrator\Desktop\Tools* directory by PE Sieve. Use PE Studio (also in the *C:\Users\Administrator\Desktop\Tools* directory) to view the dumped file inside that folder.



Run C:\Users\Administrator\Desktop\Tools\pestudio\pestudio.exe



Open XXXX.test.txt file

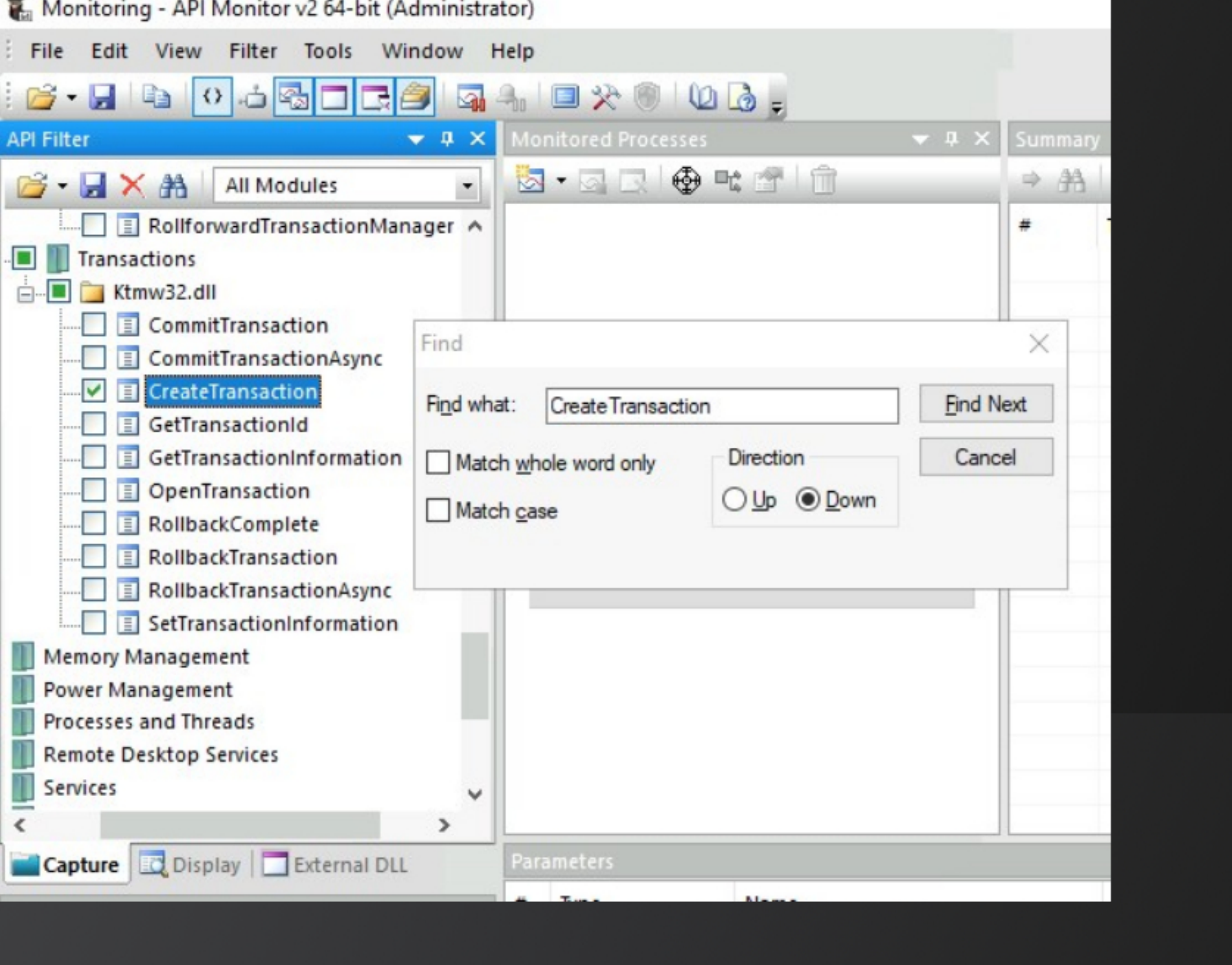


The loaded test.txt process was actually Mimikatz!

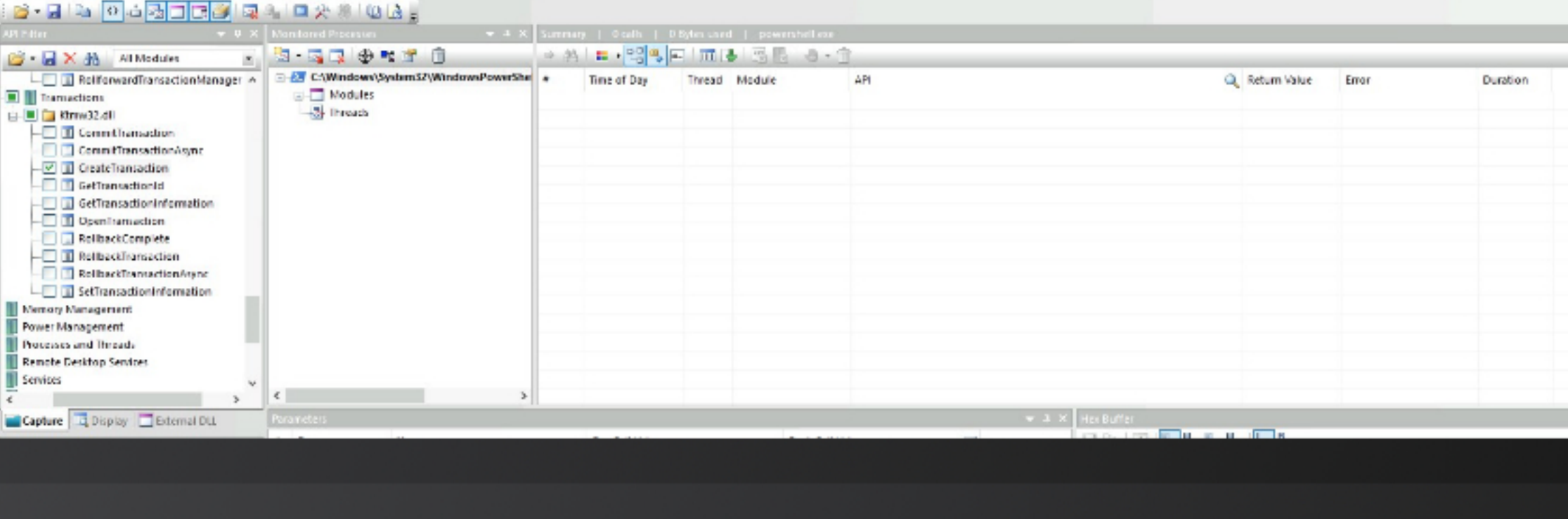
Task 2. Use API Monitor to hunt for the uncommon API call used by the Process Doppelganging technique

To hunt for an uncommon API call (such as *CreateTransaction*) that Process Doppelganging performs using API Monitor, execute the following.

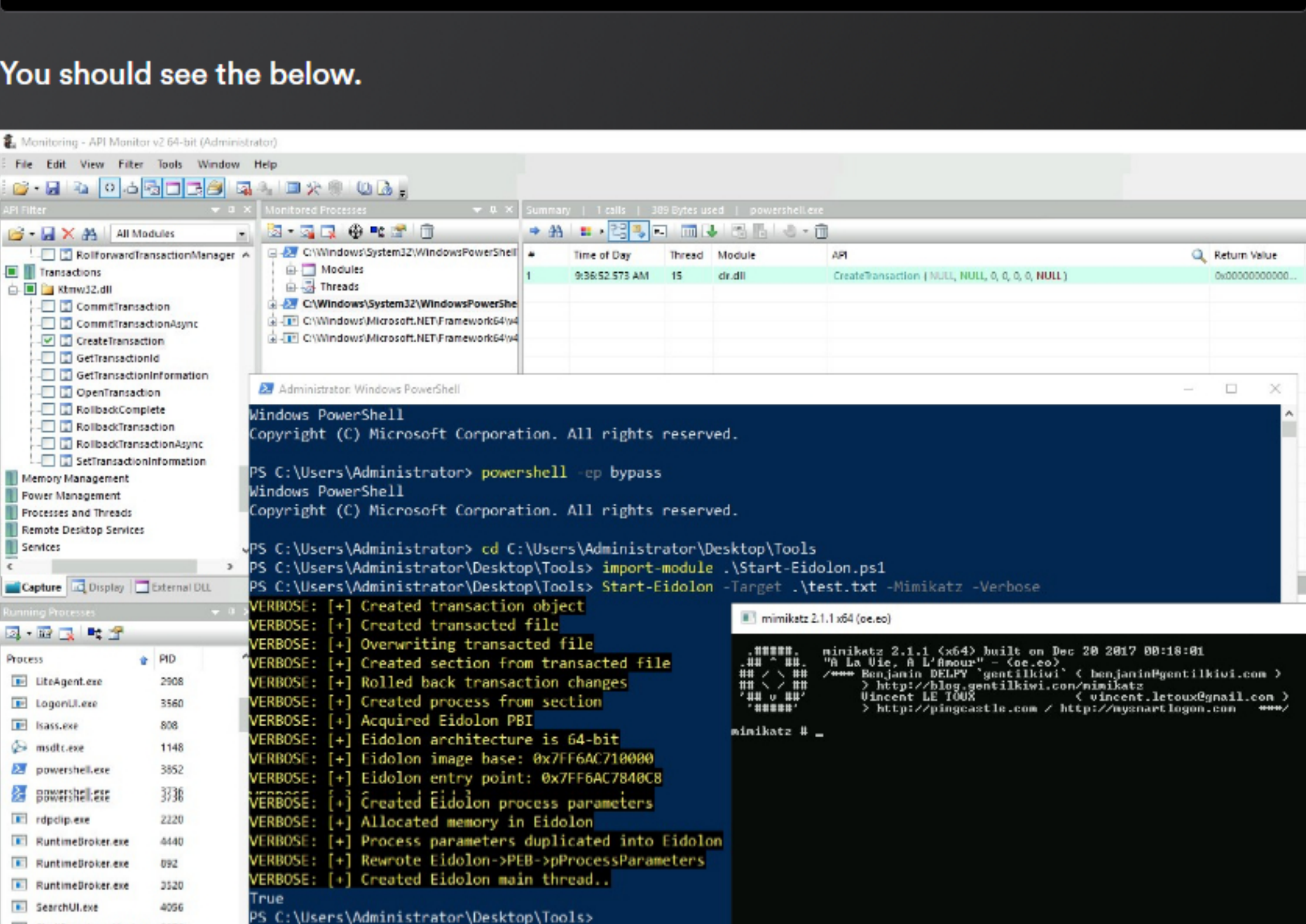
- Terminate everything you started during Task 1
- Start API Monitor x64 and search for the *CreateTransaction* API call as follows. Make sure it is ticked.



- Start *powershell.exe* as an administrator (don't click Monitor once you are prompted by API Monitor).
- Emulate the attack again. (click Monitor on the API Monitor prompt after the "powershell -ep bypass" command, we want to monitor only the second powershell process)



You should see the below.



You successfully traced the uncommon API call through Process Monitor. Nice work!

Not Running

For the best experience, choose the region closest to you. Then, start the lab to begin.

MARK UNFINISHED

Select the region closest to you

US-East

Keyboard layout: USEnglish (US)

Start lab